

PROOFLY

Evidence-First Recruitment Intelligence

“Don’t just rank candidates. Prove why.”

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1. Introduction

1.1 Purpose

PROOFLY is an AI-powered recruitment intelligence platform designed to make resume screening transparent, explainable, and evidence-driven. The system connects candidate information and job requirements to evidence extracted from the original resume.

This document describes the purpose, scope, requirements, architecture, data model, prototype workflow, security considerations, testing approach, and future enhancements of the PROOFLY prototype.

1.2 Core Philosophy

AI assists → Evidence proves → Humans decide

1.3 Product Vision

Traditional recruitment AI may provide a candidate score without showing the evidence behind it. PROOFLY aims to provide the score together with the reasoning and evidence that support the evaluation.

2. Problem Statement

Recruiters often need to evaluate many resumes against a job description. Manual screening is time-consuming, while conventional AI screening can behave like a black box.

For example, a system may show “Candidate Match: 87%” without explaining why the candidate received that score.

PROOFLY addresses this by creating an evidence chain:

Score → Job Requirement → Candidate Field → Resume Section → Exact Evidence

3. Existing System

- Traditional resume screening depends heavily on manual review.
- Some AI systems provide a match score with limited explanation.
- Missing or partially supported requirements may not be clearly separated.
- Recruiters may need to manually verify important claims in resumes.
- AI inference can be confused with confirmed resume facts.

4. Proposed System

PROOFLY introduces an evidence-first recruitment workflow that processes resumes, extracts structured information, verifies evidence, matches candidates to job requirements, calculates a PROOF SCORE™, identifies skill gaps, and generates an explainable report.

Resume → Extract → Structure → Verify → Evidence → Match → PROOF SCORE™ → Explain → Human Decision

5. Objectives

1. Automate resume processing.
2. Extract structured candidate information.
3. Preserve evidence from the original resume.
4. Match candidates against job requirements.
5. Generate an explainable PROOF SCORE™.
6. Identify missing and partially supported requirements.
7. Provide skill-gap analysis.
8. Support recruiter decision-making.
9. Generate candidate evaluation reports.
10. Keep the final hiring decision with the human recruiter.

6. Scope

6.1 In Scope

- User registration and login.
- Candidate and recruiter roles.
- Resume upload.
- PDF/DOCX processing.
- Resume field extraction.
- Evidence extraction.
- Job creation and requirements.
- Candidate applications.
- Candidate-job matching.
- PROOF SCORE™.
- Evidence Chain.
- Skill-gap analysis.
- Explainable candidate report.
- Notifications.

6.2 Prototype Limitations

- The prototype is intended for demonstration and validation.

- Production-scale deployment and enterprise integrations are outside the initial prototype.
- The system should not make an autonomous final hiring decision.

7. User Roles

7.1 Candidate

- Register and login.
- Upload and manage resumes.
- Apply for jobs.
- View application status.
- View permitted evaluation results.
- View strengths, missing requirements, and skill gaps.

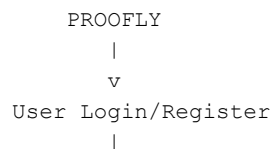
7.2 Recruiter

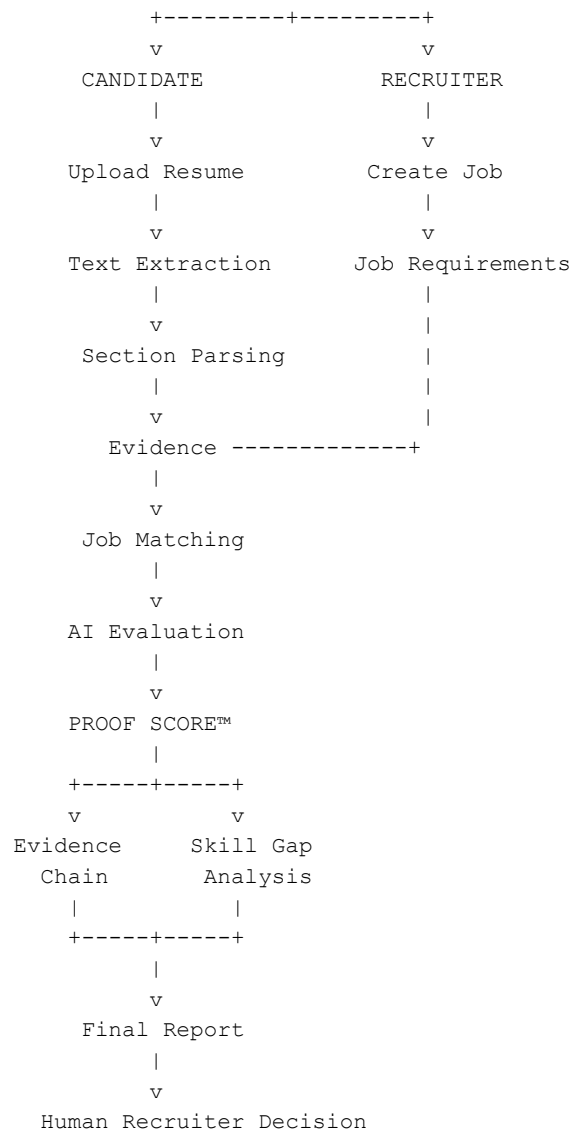
- Register and login.
- Create jobs.
- Define job requirements.
- View applications.
- Run candidate evaluation.
- View evidence and matching explanations.
- Compare candidates.
- Shortlist, reject, or mark candidates for review.
- Generate reports.

7.3 Admin

- Manage users.
- Monitor jobs and applications.
- Monitor parser/processing health.
- View failed documents and audit information.

8. System Workflow





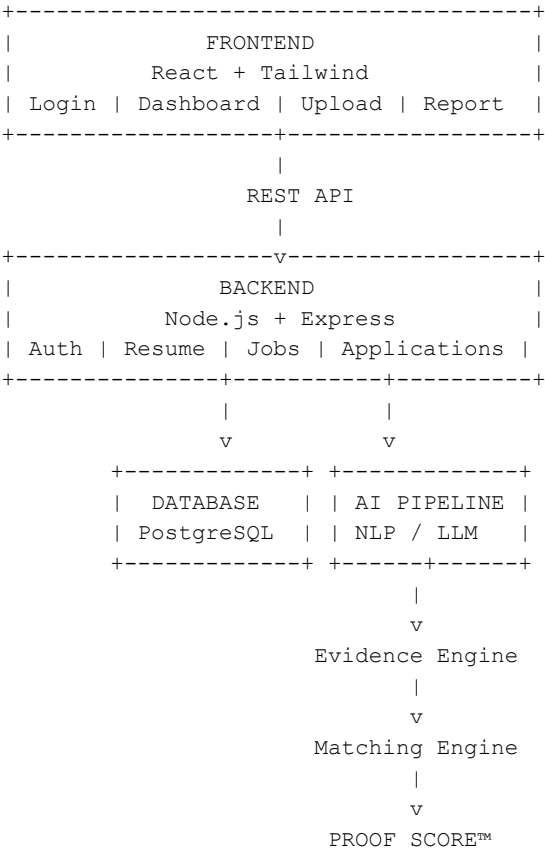
9. Functional Requirements

ID	Requirement
FR-01	System shall allow users to register.
FR-02	System shall authenticate users.
FR-03	System shall support Candidate and Recruiter roles.
FR-04	Candidate shall upload PDF/DOCX resumes.
FR-05	System shall extract resume text.
FR-06	System shall identify resume sections.
FR-07	System shall extract candidate fields.
FR-08	System shall store evidence associated with resume content.
FR-09	Recruiter shall create jobs.
FR-10	Recruiter shall define job requirements.
FR-11	Candidate shall apply for jobs.
FR-12	System shall compare candidate information with job requirements.
FR-13	System shall generate PROOF SCORE™.
FR-14	System shall show evidence supporting important matching decisions.
FR-15	System shall identify skill gaps.
FR-16	System shall generate an explainable report.
FR-17	System shall provide notifications where implemented.
FR-18	System shall restrict unauthorized information access.

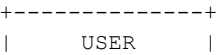
10. Non-Functional Requirements

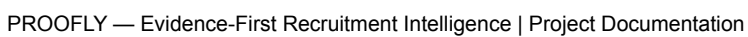
Category	Requirement
Performance	Resume processing should complete within a reasonable prototype response time.
Security	Authenticated and role-based access shall protect user data.
Reliability	Processing should produce consistent results where deterministic processing is intended.
Usability	The interface should be understandable to candidates and recruiters.
Responsiveness	The interface should support desktop and mobile layouts.
Explainability	AI inference should be distinguishable from confirmed resume evidence.
Maintainability	Modules should be separated so resume parsing, matching, scoring, and reporting can be updated independently.

11. System Architecture



12. ER Diagram





13. Database Description

Entity	Purpose
USER	Stores authentication and role information.
CANDIDATE	Stores candidate-specific information.
RECRUITER	Stores recruiter information.
RESUME	Stores uploaded resume metadata and processing status.
EVIDENCE	Stores evidence extracted from resume content.
JOB	Stores recruiter-created job information.
APPLICATION	Connects candidate, resume, and job.
FIT SCORE	Stores candidate-job matching score.
SKILL GAP	Stores missing or partially supported requirements.
FIT REPORT	Stores the explainable candidate evaluation.

14. Prototype Screens

Screen 1 — Landing Page

PROOFLY
Evidence-First Recruitment AI

“Don’t just rank candidates. Prove why.”

[Analyze Resume] [Login]

Screen 2 — Login

PROOFLY LOGIN

Email: _____
Password: _____

Role:

- o Candidate
- o Recruiter

[LOGIN]

Screen 3 — Candidate Dashboard

Welcome, Candidate

[Upload Resume]

My Applications

Python Developer	87%
Java Developer	72%

Screen 4 — Resume Upload

UPLOAD RESUME

[Choose PDF / DOCX]

[ANALYZE]

Screen 5 — Resume Analysis

RESUME ANALYSIS

Name: Candidate A
Education: B.E CSE
Skills: Python | Java | SQL
Projects: E-Waste Management

Evidence Quality: 94%

[VIEW EVIDENCE]

Screen 6 — Recruiter Dashboard

CANDIDATE	SCORE	STATUS
Candidate A	92%	Shortlist
Candidate B	87%	Review
Candidate C	64%	Review

[Compare Candidates]

Screen 7 — Job Creation

CREATE JOB

Job Title: [Python Developer]

Required:

- ☒ Python
- ☒ SQL
- ☒ FastAPI

Preferred:

- ☒ Docker
- ☒ AWS

[CREATE JOB]

Screen 8 — Match Result

MATCH SCORE

87%

[PROVE THIS DECISION]

15. Core Feature: Prove This Decision

The signature feature of PROOFLY is the ability to trace a matching score back to the evidence that supports it.

87% MATCH

|

Python Requirement
|
Candidate Skill
|
Experience Section
|
Exact Resume Evidence

The recruiter can therefore inspect why a requirement was considered matched, partially matched, or unsupported.

16. Skill Gap Analysis

SKILL GAP

STRONG

- ✓ Python
- ✓ SQL
- ✓ REST API

PARTIAL

- △ AWS

NO EVIDENCE

- × Kubernetes

“No evidence” should not be interpreted as proof that the candidate does not know the skill. It means supporting evidence was not identified in the submitted resume.

17. Technology Stack

Layer	Suggested Technology
Frontend	React + Tailwind CSS
Backend	Node.js + Express
Database	PostgreSQL
Authentication	JWT / secure password hashing
Resume Processing	PDF/DOCX text extraction
AI/NLP	NLP/LLM pipeline
API Style	REST API
Version Control	Git + GitHub

The exact technology choices can be adjusted according to the implementation team and hackathon constraints.

18. Security and Privacy

- Authentication is required for protected functions.
- Role-based authorization should restrict Candidate, Recruiter, and Admin capabilities.
- Passwords should be securely hashed rather than stored as plain text.
- Sensitive resume and candidate information should be protected.
- Recruiter-private notes and information should not be exposed to candidates.
- Secrets and API keys should not be committed to source code.
- AI-generated inference should be clearly distinguished from resume evidence.

19. Testing Plan

Test Case	Expected Result
Candidate registration	Account is created successfully.
Recruiter registration	Recruiter account is created successfully.
Login	Correct role dashboard is displayed.
Resume upload	Valid PDF/DOCX is accepted.
Invalid file	Validation error is displayed.
PDF extraction	Text is extracted.
DOCX extraction	Text is extracted.
Field extraction	Candidate fields are identified.
Evidence extraction	Evidence records are created.
Job creation	Job is saved with requirements.
Application	Candidate is linked to job.
Matching	Candidate-job evaluation is generated.
Score	PROOF SCORE™ is generated.
Evidence Chain	Score can be traced to supporting evidence.
Skill Gap	Missing/partial requirements are displayed.
Report	Explainable report is generated.
Authorization	Unauthorized access is denied.

20. Future Enhancements

- Skill Graph.
- Blind Screening.

- Job Gap Simulator.
- Candidate Comparison.
- Proof Timeline.
- Requirement DNA.
- Proof Heatmap.
- Ask PROOFLY.
- Advanced parser reliability testing.
- Enterprise HR integrations.
- Large-scale deployment.

21. Advantages

11. Explainable — recruiters can understand why a result was produced.
12. Evidence-driven — important decisions are connected to resume evidence.
13. Time-saving — reduces manual screening effort.
14. Transparent — separates evidence from AI inference.
15. Recruiter-controlled — AI assists instead of making the final hiring decision.
16. Skill-gap visibility — shows missing or partially supported requirements.
17. Prototype-friendly — the workflow can be demonstrated end-to-end.

22. Conclusion

PROOFLY transforms traditional AI resume screening into an evidence-driven decision-support process. Instead of only producing a match score, the platform connects candidate information, job requirements, evidence, matching results, skill gaps, and an explainable report.

The central message of the project is:

“The score is only the beginning. The proof is the product.”

Appendix A — Sample End-to-End Demo

18. Candidate registers and logs in.
19. Candidate uploads a sample resume.
20. System extracts resume text and candidate fields.
21. Recruiter creates a Python Developer job.
22. Recruiter defines Python, SQL, FastAPI, Docker, and AWS requirements.

23. Candidate applies for the job.
24. System evaluates the candidate.
25. System generates a sample PROOF SCORE™ such as 87%.
26. Recruiter clicks “Prove This Decision”.
27. System displays evidence for matched and partially matched requirements.
28. System displays skill gaps and no-evidence requirements.
29. Recruiter generates the final candidate report.

Appendix B — Glossary

Term	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ATS	Applicant Tracking System
Evidence Chain	Trace connecting a matching result to source resume evidence.
PROOF SCORE™	Explainable candidate-job matching score used by the PROOFLY concept.
Skill Gap	A requirement that is missing, partially supported, or lacks evidence in the submitted resume.
Candidate	Person applying for a job.
Recruiter	User who creates jobs and evaluates candidates.
Admin	User responsible for platform-level management and monitoring.

Appendix C — References

1. Project handout/build specification supplied for PROOFLY: Evidence-First Recruitment Intelligence.
2. University of Central Florida, Software Requirements Specification Template — used only as a general reference for organizing requirements documentation.
3. Michigan State University, Software Requirements Specification template — used only as a general reference for SRS organization.